

Simple Machines and Mechanical Advantage

Applying work and energy principles to the mechanical devices that have shaped human civilization.

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Opening Question: Why does a long crowbar pry open a crate that your bare hands cannot budge? Why does a ramp let you roll a heavy barrel into a truck that you could never lift straight up? Why does a block-and-tackle arrangement let a single person hoist a piano that would require ten people pulling directly? In every case, a machine is not producing extra energy — it is redistributing the energy you put in, trading distance for force or force for distance. Understanding how and why this works is the subject of this chapter.

18.1 Why Machines Matter

Every tool humans have invented to ease physical work is, at its core, a device for redirecting or redistributing force. A crowbar, a doorknob, a screw, a set of stairs, a pulley hoisting an anchor — all of these are machines in the physical sense. They do not create energy. They do not violate the law of conservation of energy. What they do is allow a person to apply a smaller force over a larger distance to accomplish work that would otherwise require a larger force over a smaller distance.

This chapter closes Part III by showing that conservation of energy is not just an abstract principle — it is the governing constraint behind every mechanical device ever built. The work you put into a machine equals the work the machine delivers (minus losses to friction and deformation). From that single constraint, the entire theory of mechanical advantage follows.

The six classical simple machines — the lever, the pulley, the inclined plane, the wedge, the screw, and the wheel and axle — are the building blocks of all mechanical technology. A bicycle combines wheels, axles, and lever-like crank systems. A car jack is a screw. An escalator is an inclined plane that moves. A compound bow applies multiple lever and pulley principles simultaneously. The physics of every one of these devices flows from the ideas developed here.

Bridge Note: In the MGU Physics advanced series (Volume I), machine efficiency is revisited through the Lagrangian formalism, where constraints on mechanical systems are expressed as generalized coordinates. The intuition you build here — that machines trade force for distance subject to energy conservation — becomes the physical motivation for why constrained systems have fewer degrees of freedom.

18.2 What Makes a Machine "Simple"

The six simple machines are the elementary force-redirecting devices. They are called "simple" not because they are trivial, but because they are the irreducible building blocks from which all more complex machines are constructed. Each one has a single input force and a single output force.

Definition: A **simple machine** is a device that changes the magnitude, direction, or both, of an applied force through a single mechanical action, operating on the principle that the work input equals the work output in an ideal (frictionless) system.

The six simple machines are:

1. **Lever** — a rigid bar that rotates about a fixed pivot (fulcrum)
2. **Pulley** — a wheel with a grooved rim, used to redirect or multiply force via a rope or cable
3. **Inclined plane** — a flat surface set at an angle to the horizontal
4. **Wedge** — two inclined planes joined back-to-back, converting a driving force into splitting forces
5. **Screw** — an inclined plane wrapped helically around a cylinder
6. **Wheel and axle** — a large wheel fixed to a smaller axle, rotating as a unit

Each of these will be analyzed in detail in Sections 18.8 through 18.12. Before examining them individually, we need the general framework that applies to all of them: the concept of mechanical advantage and the role of energy conservation.

18.3 Input Force, Output Force, and Work

Every machine has an input side and an output side. The **input force** F_{in} (also called the effort force) is the force you apply to the machine. The **output force** F_{out} (also called the load force or resistance force) is the force the machine exerts on the object being moved.

The machine also involves two displacements: the **input displacement** d_{in} through which the input force acts, and the **output displacement** d_{out} through which the output force acts.

The work done on the machine by the effort and the work done by the machine on the load are:

$$W_{in} = F_{in} \cdot d_{in}$$

$$W_{out} = F_{out} \cdot d_{out}$$

For an **ideal machine** (no friction, no deformation, no internal energy losses), conservation of energy requires:

$$W_{in} = W_{out}$$

$$F_{in} \cdot d_{in} = F_{out} \cdot d_{out}$$

This single equation encodes the fundamental trade-off of every machine: if the output force is larger than the input force ($F_{out} > F_{in}$), then the output displacement must be smaller than the input displacement ($d_{out} < d_{in}$) by the same factor. You gain force, but you must move through a greater distance to do it.

Key Point: A machine cannot create energy. If a machine multiplies force by a factor of 5, it must also reduce displacement by a factor of 5. The total work is unchanged. Machines trade force for distance — never for energy.

FIGURE 18.1: Schematic of a general machine showing input force F_{in} , input displacement d_{in} , output force F_{out} , and output displacement d_{out} , with the energy conservation constraint $F_{in} d_{in} = F_{out} d_{out}$.

18.4 Mechanical Advantage

The **mechanical advantage** (MA) of a machine is the ratio of the output force to the input force:

$$MA = F_{out}/F_{in}$$

Mechanical advantage is dimensionless — it is a pure ratio of forces. It tells you how many times the machine multiplies the input force.

- If **MA** $>$ **1**: the machine multiplies force. You exert a smaller force and the machine exerts a larger force on the load. (You must move through a greater distance to compensate.)

- If **MA = 1**: the machine does not multiply force; it only changes direction.
- If **MA < 1**: the machine reduces force but multiplies displacement. (Used when increased speed or range of motion is the goal, such as in a baseball bat or a fishing rod.)

MA Value	Effect	Example
MA > 1	Multiplies force; reduces displacement	Crowbar, hydraulic jack, block and tackle
MA = 1	Redirects force only	Single fixed pulley
MA < 1	Reduces force but multiplies speed/displacement	Baseball bat, fishing rod, tweezers

Table 18.1:

Common Mistake: Students sometimes think a mechanical advantage greater than 1 means the machine does more work. It does not. The work output equals the work input (for an ideal machine). A higher mechanical advantage means the same work is delivered with less force over more distance — not more work overall.

18.5 Ideal Mechanical Advantage

The **ideal mechanical advantage** (IMA) is the mechanical advantage of a perfect machine with no friction or internal energy losses. It can always be calculated from the geometry of the machine alone — from the ratio of input to output displacements:

$$\text{IMA} = d_{\text{in}}/d_{\text{out}}$$

This follows directly from the ideal work equation: $F_{\text{in}} d_{\text{in}} = F_{\text{out}} d_{\text{out}}$, which gives $F_{\text{out}}/F_{\text{in}} = d_{\text{in}}/d_{\text{out}}$.

The IMA is a geometric property of the machine. It does not depend on the forces applied or on whether friction is present. You can compute the IMA from a ruler and the machine's dimensions alone.

Example — Lever IMA: A crowbar has its fulcrum 0.10 m from the load end and 0.90 m from the effort end. When the effort end moves down by 9.0 cm, the load end moves up by 1.0 cm. $\text{IMA} = d_{\text{in}}/d_{\text{out}} = 9.0 \text{ cm} / 1.0 \text{ cm} = 9$. This lever can theoretically produce 9 times the input force at the output.

Example — Inclined Plane IMA: A ramp is 4.0 m long and rises 0.80 m. To move a box along the ramp (input displacement = 4.0 m), the box rises 0.80 m vertically (output displacement = 0.80 m in the direction of the lifted weight). $\text{IMA} = 4.0 / 0.80 = 5$.

For each simple machine, the IMA has a simple geometric formula derived from the machine's dimensions. These are developed in Sections 18.8–18.12.

18.6 Actual Mechanical Advantage

In any real machine, friction converts some of the input work to thermal energy. As a result, the output force is always somewhat less than the ideal prediction. The **actual mechanical advantage** (AMA) is the ratio of the measured output force to the measured input force:

$$\text{AMA} = F_{\text{out,measured}}/F_{\text{in,measured}}$$

The AMA is always less than or equal to the IMA:

$$\text{AMA} \leq \text{IMA}$$

The difference between IMA and AMA reflects energy lost to friction. The ratio AMA/IMA is the efficiency of the machine.

Key Point: The IMA is calculated from geometry and tells you what a perfect machine would achieve. The AMA is measured from actual forces and tells you what the real machine achieves. The IMA is always the upper limit; friction always brings AMA below IMA.

Worked Example 18.1 — Finding AMA

A person exerts 200 N of input force on a lever to lift a 900 N load.

Given: $F_{in} = 200 \text{ N}$, $F_{out} = 900 \text{ N}$.

Find: AMA.

$$AMA = F_{out}/F_{in} = 900 \text{ N}/200 \text{ N} = 4.5$$

The lever multiplies the input force by a factor of 4.5. In other words, 200 N of effort produces 900 N of lifting force.

18.7 Efficiency in Real Machines

The **efficiency** of a machine measures how much of the input work is converted to useful output work:

$$\text{efficiency} = W_{out}/W_{in} \times 100\%$$

Equivalently:

$$\text{efficiency} = AMA/IMA \times 100\%$$

Efficiency is always between 0% (all energy lost, no useful output) and 100% (perfect machine, no losses). No real machine reaches 100%, although well-designed machines with rolling bearings and lubrication can approach 95–99% efficiency.

The energy not delivered as useful output work becomes thermal energy in the machine's components — heating the axle, the rope, the surfaces sliding against each other.

$$W_{in} = W_{out} + W_{friction}$$

$$W_{friction} = W_{in}(1 - \text{efficiency})$$

Worked Example 18.2 — Calculating Efficiency

A pulley system requires 60.0 N of input force applied over 3.0 m to lift a 150 N load by 1.0 m.

Step 1 — Compute work in and work out:

$$W_{in} = F_{in} \cdot d_{in} = 60.0 \text{ N} \times 3.0 \text{ m} = 180 \text{ J}$$

$$W_{out} = F_{out} \cdot d_{out} = 150 \text{ N} \times 1.0 \text{ m} = 150 \text{ J}$$

Step 2 — Efficiency:

$$\text{efficiency} = 150 \text{ J}/180 \text{ J} \times 100\% = 83.3\%$$

Step 3 — Energy lost to friction:

$$W_{friction} = 180 \text{ J} - 150 \text{ J} = 30 \text{ J}$$

Step 4 — Find IMA and AMA:

$$\text{IMA} = d_{\text{in}}/d_{\text{out}} = 3.0 \text{ m}/1.0 \text{ m} = 3.0$$

$$\text{AMA} = F_{\text{out}}/F_{\text{in}} = 150 \text{ N}/60.0 \text{ N} = 2.5$$

Check: $\text{efficiency} = \text{AMA}/\text{IMA} = 2.5/3.0 = 83.3\%$ ✓

Machine Type	Typical Efficiency
Ball-bearing wheel and axle	95–99%
Well-lubricated gear system	90–97%
Block and tackle (rope pulley)	70–90%
Inclined plane (wood on wood)	50–80%
Screw jack	15–50%
Human muscle (mechanical efficiency)	20–30%

Table 18.2:

Historical Note: The study of efficiency in machines was one of the driving forces behind the development of thermodynamics in the nineteenth century. Engineers building steam engines needed to know why no machine could convert heat entirely to work — a question that eventually led to the second law of thermodynamics and the concept of entropy. The simple machines analyzed in this chapter are the classical starting point for that inquiry.

18.8 Levers

A **lever** is a rigid bar that pivots about a fixed point called the **fulcrum**. Forces applied to either side of the fulcrum cause rotation. The three quantities that define a lever's behavior are:

- The **effort force** F_E and its distance from the fulcrum, called the **effort arm** d_E
- The **load force** F_L and its distance from the fulcrum, called the **load arm** d_L

For an ideal lever in equilibrium (no acceleration), the principle of moments (introduced formally in Chapter 20) gives:

$$F_E \cdot d_E = F_L \cdot d_L$$

The IMA of a lever is:

$$\text{IMA} = d_E/d_L = F_L/F_E$$

A long effort arm and a short load arm give a large IMA. To lift a heavy load with little effort, position the fulcrum close to the load and far from where you push.

Classes of Levers

Levers are classified by the relative positions of the fulcrum (F), the effort force (E), and the load (L):

Class	Arrangement	IMA	Examples
First class	Fulcrum between E and L	Can be any value	Seesaw, crowbar, scissors
Second class	Load between F and E	Always > 1	Wheelbarrow, nutcracker, door
Third class	Effort between F and L	Always < 1	Tweezers, fishing rod, forearm

Table 18.3:

FIGURE 18.2: Diagrams of all three lever classes showing the relative positions of fulcrum, effort, and load, with labeled effort arm and load arm for each.

First-class levers have the fulcrum between the effort and the load, like a seesaw or a crowbar. The effort and load move in opposite directions. The IMA can be greater than, equal to, or less than 1 depending on the arm lengths.

Second-class levers have the load between the fulcrum and the effort. The wheelbarrow is the classic example: the wheel is the fulcrum, the load sits in the bin between the wheel and your hands, and you lift at the handles (the far end). The effort and load move in the same direction, and the IMA is always greater than 1.

Third-class levers have the effort between the fulcrum and the load. The forearm is a biological third-class lever: the elbow is the fulcrum, the bicep applies effort close to the elbow, and the hand holds the load far away. The IMA is always less than 1 — you exert more force than you receive — but you gain speed and range of motion. This is why the human arm is not well-designed for heavy lifting, but is excellent for throwing and swinging.

Worked Example 18.3 — Crowbar as a First-Class Lever

A crowbar is 1.50 m long. The fulcrum is placed 0.10 m from the pry end (load side). A worker pushes down on the far end.

Step 1 — Identify arm lengths:

- Load arm $d_L = 0.10$ m
- Effort arm $d_E = 1.50 - 0.10 = 1.40$ m

Step 2 — IMA:

$$\text{IMA} = d_E/d_L = 1.40 \text{ m}/0.10 \text{ m} = 14$$

Step 3 — If the worker pushes with 80 N, what force does the crowbar exert on the load (ideal)?

$$F_L = \text{IMA} \times F_E = 14 \times 80 \text{ N} = 1120 \text{ N}$$

This 80 N effort produces over 1 kN of force at the load end — more than enough to pry open a nail.

Key Point: The closer the fulcrum is to the load, the greater the mechanical advantage. This is why you position a crowbar's fulcrum as close to the stuck nail as possible.

18.9 Pulleys

A **pulley** is a wheel with a grooved rim that a rope, cable, or chain runs along. Pulleys serve two purposes: they can change the direction of a force (a single fixed pulley) or they can multiply force (a movable pulley or a block-and-tackle system).

Fixed Pulley

A **fixed pulley** is attached to a fixed support. The rope runs over it, and you pull one side to lift the other. The fixed pulley does not multiply force: both sides of the rope carry the same tension, so $F_{\text{out}} = F_{\text{in}}$. The IMA of a single fixed pulley is 1.

Its only advantage is direction: you can pull down (using your weight to assist) to lift a load up.

Movable Pulley

A **movable pulley** is attached to the load rather than to a fixed point. When you pull the rope, two segments of rope support the load, so the load is supported by two tension forces while you apply only one. The IMA of a single movable pulley is 2.

$$\text{IMA (single movable pulley)} = 2$$

The trade-off: you must pull the rope twice as far as the load rises.

Block-and-Tackle Systems

A **block-and-tackle** combines multiple fixed and movable pulleys. The IMA equals the number of rope segments that directly support the movable block:

$$\text{IMA (block-and-tackle)} = n$$

where n is the number of rope segments supporting the movable block.

FIGURE 18.3: Diagrams of (a) a single fixed pulley ($\text{IMA} = 1$), (b) a single movable pulley ($\text{IMA} = 2$), and (c) a four-segment block-and-tackle system ($\text{IMA} = 4$), with rope segments labeled.

Pulley System	Rope Segments Supporting Load	IMA
Single fixed pulley	1	1
Single movable pulley	2	2
Two movable pulleys	4	4
Three movable pulleys	6	6

Table 18.4:

Worked Example 18.4 — Block-and-Tackle

A block-and-tackle system has 4 rope segments supporting the movable block. A worker needs to lift a 600 N load. Assume efficiency of 80%.

Step 1 — IMA: $\text{IMA} = 4$

Step 2 — Ideal effort required:

$$F_{\text{in, ideal}} = F_{\text{out}}/\text{IMA} = 600 \text{ N}/4 = 150 \text{ N}$$

Step 3 — Actual effort required (accounting for 80% efficiency):

$$W_{\text{in}} = W_{\text{out}}/\text{efficiency} = 600 \times d_{\text{out}}/0.80$$

Since efficiency = AMA/IMA :

$$\text{AMA} = \text{efficiency} \times \text{IMA} = 0.80 \times 4 = 3.2$$

$$F_{\text{in, actual}} = F_{\text{out}}/\text{AMA} = 600 \text{ N}/3.2 = 187.5 \text{ N}$$

The worker must exert approximately 188 N instead of the ideal 150 N, with the extra 37.5 N of effort compensating for friction in the pulley system.

18.10 Inclined Planes

An **inclined plane** is a flat surface tilted at an angle θ to the horizontal. It allows a load to be moved to a height h by pushing it along a longer distance L (the length of the ramp surface). The vertical height risen is the useful output displacement; the ramp length is the input displacement.

$$\text{IMA} = L/h = 1/\sin\theta$$

Because $L > h$ for any angle $\theta < 90^\circ$, the IMA is always greater than 1. The shallower the ramp, the greater the mechanical advantage.

FIGURE 18.4: Inclined plane diagram showing ramp length L , vertical height h , angle θ , and the forces on a box being pushed up the ramp.

Connection to Chapter 11: This is the same geometry encountered when resolving gravity into components on an inclined surface. The component of gravity along the ramp is $mg\sin\theta$, and the component perpendicular to the ramp is $mg\cos\theta$. The force needed to push a frictionless box up the ramp equals $mg\sin\theta = mg \cdot (h/L)$. Notice:

$$F_{in}, ideal F_{out} = mg\sin\theta/mg = \sin\theta = h/L = 1/IMA$$

This confirms the energy conservation derivation: $IMA = L/h$.

Worked Example 18.5 — Ramp Loading a Truck

A loading ramp is 4.50 m long and rises 1.20 m. A 500 kg barrel needs to be rolled into the truck. Assume ideal (no friction).

Step 1 — IMA:

$$IMA = L/h = 4.50 \text{ m} / 1.20 \text{ m} = 3.75$$

Step 2 — Weight of barrel:

$$W = mg = 500 \times 9.81 = 4905 \text{ N}$$

Step 3 — Required effort (ideal):

$$F_{in} = F_{out}/IMA = 4905 \text{ N} / 3.75 = 1308 \text{ N} \approx 1310 \text{ N}$$

Step 4 — Reality check: with friction ($\mu_k = 0.20$):

$$\text{Normal force on ramp: } N = mg\cos\theta = 4905 \times \cos(15.5^\circ) = 4905 \times 0.965 = 4733 \text{ N}$$

$$\text{Friction force: } f_k = \mu_k N = 0.20 \times 4733 = 947 \text{ N}$$

$$\text{Total effort required: } F_{in} = 1308 + 947 = 2255 \text{ N}$$

$$\text{Efficiency: } \text{efficiency} = 1308/2255 \times 100\% = 58\%$$

Common Mistake: Students often forget that friction on a ramp acts along the ramp surface (opposing the direction of motion), not vertically. The friction force uses the normal force $N = mg\cos\theta$, not the full weight mg . On a 30° ramp, $N = mg\cos 30^\circ = 0.866 mg$ — less than the full weight.

18.11 Wedges and Screws

Wedges

A **wedge** is two inclined planes joined back-to-back. When driven forward, its sloping sides convert the input force (directed along the wedge's axis) into spreading forces perpendicular to the wedge surfaces. Axes, chisels, knives, nails, and door stops are all wedges.

The IMA of a wedge depends on its geometry:

$$IMA = L/w$$

where L is the length of the wedge and w is its maximum width (or the width at the thick end). A longer, thinner wedge has a higher IMA and splits material more easily for a given driving force.

FIGURE 18.5: Diagram of a wedge showing length L , width w , input force (driving the wedge), and output forces (perpendicular splitting forces on either side).

In practice, wedges experience enormous friction because the wedge surfaces are pressed hard against the material being split. This is why wedge efficiency is often low (30–60%), and also why

wedges are self-locking: friction keeps them in place once driven in, because the driving force is not needed to hold them there.

Screws

A **screw** is an inclined plane wrapped helically around a cylinder. When the screw makes one full rotation, it advances by a distance called the **pitch *p*** (the distance between adjacent threads). The effort is applied at radius ***r*** (the length of the screwdriver handle or wrench arm), and the screw advances ***p*** in each full rotation.

$$IMA = 2\pi \, r/p$$

The effort moves through the circumference of a circle (**$2\pi \, r$**) for each rotation, while the screw advances only the pitch distance ***p***. Because ***r*** ***p*** for most screws, the IMA is very large — often 50 to 200 or more.

Example: A bolt has a pitch of 1.0 mm (0.001 m) and is tightened with a wrench of arm length 15 cm (0.15 m).

$$IMA = 2\pi \times 0.15/0.001 = 0.9425/0.001 = 942.5$$

This enormous mechanical advantage allows a hand-tightened bolt to exert thousands of newtons of clamping force. However, as with wedges, friction is very significant in screws, and actual efficiency is typically 15–50%. The low efficiency of a screw is also what makes it self-locking: the load cannot back-drive the screw because friction holds it in place.

Key Point: Self-locking is a desirable feature of wedges and screws precisely because of their low efficiency. A bolt that could back-drive (loosen) under load would be useless. The "wasted" energy from friction keeps the fastener tight.

18.12 Wheels and Axles

A **wheel and axle** consists of a large wheel (radius ***R***) rigidly attached to a smaller axle (radius ***r***), rotating as a unit around a common axis. Effort is applied at the rim of the wheel; the axle delivers the output force to the load.

$$IMA = R/r$$

When effort rotates the wheel through one full turn, both the wheel rim and the axle rim travel the same angular distance (**2π radians**), but different linear distances: the wheel rim travels **$2\pi \, R$** and the axle rim travels **$2\pi \, r$** . Effort applied at the rim of the large wheel thus acts over a longer distance (**$2\pi \, R$**) than the axle moves (**$2\pi \, r$**), giving a mechanical advantage of **R/r** .

FIGURE 18.6: Diagram of a wheel and axle showing the large wheel of radius *R*, the small axle of radius *r*, the effort applied at the wheel rim, and the load suspended from the axle.

Wheel-and-Axle Application	Wheel Radius <i>R</i>	Axle Radius <i>r</i>	IMA
Doorknob (handle to shaft)	~3.5 cm	~0.5 cm	~7
Steering wheel (wheel to column)	~18 cm	~1.5 cm	~12
Winch (crank radius to drum radius)	~30 cm	~5 cm	~6
Screwdriver (handle to blade)	~2.0 cm	~0.3 cm	~7

Table 18.5:

Gears are a variation on the wheel-and-axle principle. Two meshed gears of different sizes form a wheel-and-axle pair where the circumferences (and therefore the number of teeth) determine the mechanical advantage. A small gear driving a large gear multiplies torque (force) but reduces rotational speed; a large gear driving a small gear multiplies speed but reduces torque.

Bridge Note: In Chapter 19, the wheel-and-axle analysis becomes the foundation for understanding torque — the rotational equivalent of force. The product $\mathbf{F} \times \mathbf{R}$ for the wheel and $\mathbf{F}_{\text{out}} \times \mathbf{r}$ for the axle are both torques, and the equilibrium condition $\mathbf{F} \times \mathbf{R} = \mathbf{F}_{\text{out}} \times \mathbf{r}$ is the rotational equilibrium condition = 0 applied to the wheel-and-axle system.

18.13 Compound Machines

A **compound machine** is a combination of two or more simple machines working together. The overall mechanical advantage of a compound machine is the product of the individual mechanical advantages of its components:

$$MA_{\text{total}} = MA_1 \times MA_2 \times MA_3 \times \dots$$

This multiplication of mechanical advantages is why compound machines can achieve very large force multiplication. A bicycle, for example, combines:

- A wheel and axle (the pedal crank and chainring gear)
- Another wheel and axle (the rear sprocket and wheel)
- Levers (the brakes, the handlebars)

The overall gear ratio of a bicycle is the product of the chainring-to-sprocket gear ratio and the wheel-to-ground relationship.

A **car jack** is a screw (very high IMA) often combined with a lever mechanism (additional IMA) to allow one person to lift several tons of vehicle with a modest hand force.

Example — Compound Machine IMA: A screwdriver (IMA = 7) is used to tighten a screw (IMA = 100). Overall IMA = $7 \times 100 = 700$. This means the hand force is multiplied 700 times at the point where the screw tip contacts the material being fastened.

The efficiency of a compound machine is the product of the efficiencies of its components:

$$\text{efficiency}_{\text{total}} = e_1 \times e_2 \times e_3 \times \dots$$

This multiplication of efficiencies means compound machines are generally less efficient than their individual components. Four components each at 90% efficiency give an overall efficiency of $0.90^4 = 0.66 = 66\%$. This is why complex machinery requires active lubrication to maintain acceptable efficiency.

18.14 Common Mistakes in Mechanical Advantage Problems

Mistake 1: Confusing IMA and AMA

The IMA is a geometric property calculated from distances alone. The AMA is measured from actual forces and is always less than or equal to the IMA (because friction reduces the output force). Never use $d_{\text{in}}/d_{\text{out}}$ and call it the AMA — that is the IMA. Measure forces to find AMA.

Mistake 2: Thinking a machine creates energy

A machine with $IMA = 10$ does not produce 10 times as much work as you put in. It produces the same work (minus losses), delivered at 10 times the force over $1/10$ the distance. Always check: $W_{in} \geq W_{out}$. If your calculation implies $W_{out} > W_{in}$, something is wrong.

Mistake 3: Ignoring direction in pulley problems

When counting rope segments to find the IMA of a block-and-tackle, count only the segments attached to the movable block. The segment attached to the pull end (the effort end) does not count — it is the effort segment, not a support segment.

Mistake 4: Using the wrong reference length for screw IMA

For a screw or bolt tightened with a wrench, the radius r in $IMA = 2\pi r/p$ is the distance from the axis of the screw to where the force is applied — the length of the wrench arm, not the radius of the screw's shaft. A longer wrench gives greater mechanical advantage on the same bolt.

Mistake 5: Forgetting that efficiency reduces AMA, not IMA

If a machine has $IMA = 6$ and efficiency = 75%, the AMA is $0.75 \times 6 = 4.5$, not 6. The input displacement does not change with friction; the output force changes. Efficiency losses reduce how much force the machine delivers, not the geometric relationship between input and output distances.

Mistake 6: Misidentifying the class of a lever

Determine which of the three positions (fulcrum, effort, load) is in the middle. Whichever point is between the other two determines the class. Fulcrum in the middle: first class. Load in the middle: second class. Effort in the middle: third class. Sketch the lever if you are unsure.

Common Mistake: A common error is to assume that all levers with $MA > 1$ are first-class levers. A wheelbarrow (second-class) and a nutcracker (second-class) both have $MA > 1$, but the fulcrum is at one end. Draw the lever and identify all three points before classifying.

Chapter Summary

Machines redistribute forces subject to the constraint of energy conservation. The total work output of an ideal machine equals the total work input. Real machines have lower output work due to friction.

Core relationships:

$$MA = F_{out}/F_{in} \quad IMA = d_{in}/d_{out} \quad \text{efficiency} = W_{out}/W_{in} \times 100\% = AMA/IMA \times 100\%$$

- **Machines do not create energy.** They trade force for distance, keeping total work constant in an ideal system.
- **IMA** is geometric: calculated from the dimensions of the machine alone, using d_{in}/d_{out} .
- **AMA** is measured: the ratio of actual output force to actual input force; always $\leq IMA$.
- **Efficiency** is always $\leq 100\%$; friction converts unused input work to thermal energy.
- **Lever IMA** = effort arm / load arm = d_E / d_L . Three classes: fulcrum between E and L (1st), load between F and E (2nd), effort between F and L (3rd).
- **Pulley IMA** = number of rope segments supporting the movable block.
- **Inclined plane IMA** = ramp length / vertical height = $L/h = 1/\sin\theta$.
- **Screw IMA** = $2\pi r / p$ where r is the wrench arm length and p is the pitch.
- **Wheel and axle IMA** = wheel radius / axle radius = R/r .

- **Compound machine:** total MA = product of individual MAs; total efficiency = product of individual efficiencies.

Key Terms

simple machine: A device that changes the magnitude, direction, or both, of an applied force using a single mechanical action. The six simple machines are: lever, pulley, inclined plane, wedge, screw, and wheel and axle.

mechanical advantage (MA): The ratio of the output force to the input force: $MA = F_{\text{out}}/F_{\text{in}}$. Dimensionless.

ideal mechanical advantage (IMA): The mechanical advantage of a frictionless machine, calculated from geometry alone: $IMA = d_{\text{in}}/d_{\text{out}}$.

actual mechanical advantage (AMA): The measured mechanical advantage of a real machine, reduced below the IMA by friction: $AMA = F_{\text{out, measured}}/F_{\text{in, measured}}$.

efficiency: The ratio of useful work output to total work input, expressed as a percentage: $\text{efficiency} = (W_{\text{out}}/W_{\text{in}}) \times 100\%$. Always $\leq 100\%$ for real machines.

effort force (FE): The force applied to the machine by the user; the input force.

load force (FL): The force the machine applies to the object being moved; the output force.

fulcrum: The fixed pivot point of a lever.

effort arm (dE): The perpendicular distance from the fulcrum to the line of action of the effort force in a lever.

load arm (dL): The perpendicular distance from the fulcrum to the line of action of the load force in a lever.

pitch: The distance between adjacent threads of a screw; the distance the screw advances per full rotation.

block and tackle: A pulley system combining multiple fixed and movable pulleys to achieve a large mechanical advantage equal to the number of rope segments supporting the movable block.

compound machine: A combination of two or more simple machines; its total MA is the product of the individual MAs.

self-locking: The property of a wedge or screw whereby friction prevents the load from back-driving the machine; a consequence of low efficiency.

Concept Review Questions

Answer in complete sentences. No calculation required.

7. What is the fundamental trade-off that every machine makes? Explain why a machine with $MA > 1$ does not violate conservation of energy.
8. Explain the difference between IMA and AMA. Which is always larger? Why?
9. A pulley system has an IMA of 4 but an actual efficiency of 70%. Calculate the AMA. Where does the remaining 30% of the input energy go?
10. Describe the three classes of levers. Give one everyday example of each. For each class, state whether the MA is always greater than, less than, or either greater or less than 1.

11. Why does a longer screwdriver handle make it easier to tighten a screw? Which quantity — IMA or AMA — are you increasing by using a longer handle?
12. A wedge and a screw are both examples of inclined planes. Explain how each one converts an input force into an output force, and in what direction the output force acts relative to the input force.
13. Why is the efficiency of a screw typically much lower than the efficiency of a ball-bearing wheel and axle? Why is the screw's low efficiency often considered an advantage rather than a disadvantage?
14. A student claims that a compound machine with five components, each of $MA = 3$, has a total MA of 15. Another student says the MA is $3^5 = 243$. Who is correct? Explain.
15. Explain why a thin, long wedge splits wood more easily than a short, wide wedge of the same material, assuming the same driving force.
16. Give one example from everyday life where $MA < 1$ is desirable. What is gained by using a machine with $MA < 1$?

Mechanical Advantage Problems

Show all work. Include units in every step.

17. A lever has an effort arm of 2.40 m and a load arm of 0.30 m. **(a)** What is the IMA? **(b)** If an effort force of 50.0 N is applied, what is the ideal output force? **(c)** If the actual output force is 320 N, what is the AMA? **(d)** What is the efficiency?
18. A block-and-tackle system has 5 rope segments supporting the movable block. A worker applies 80.0 N to lift a load. **(a)** What is the IMA? **(b)** What is the ideal weight of the load the system can lift? **(c)** If the efficiency is 78%, what is the actual weight of the load lifted?
19. A ramp is 6.00 m long and rises 1.50 m. **(a)** Find the IMA. **(b)** What force is needed to push a frictionless 200 kg crate up the ramp? **(c)** If friction requires an additional 180 N along the ramp, what is the actual input force? **(d)** Calculate the efficiency.
20. A wheel-and-axle system has a wheel radius of 25.0 cm and an axle radius of 4.0 cm. **(a)** Find the IMA. **(b)** What effort is needed to lift a 900 N load (ideal)? **(c)** If the efficiency is 88%, what effort is actually required?
21. A screw has a pitch of 2.0 mm and is tightened with a wrench whose handle is 18.0 cm long. **(a)** Find the IMA. **(b)** If the applied torque (hand force $\times$ handle length) is 15.0 N·m, what clamping force does the screw exert (ideal)? **(c)** If the actual efficiency is 30%, what is the actual clamping force?
22. A first-class lever has a load of 500 N located 0.20 m from the fulcrum. Where must the effort be applied to achieve an IMA of 8?
23. A person uses a wheel and axle to lift a 600 N load. The wheel has radius 30.0 cm and the axle has radius 5.0 cm. The efficiency is 85%. **(a)** Find AMA. **(b)** Find the required effort force. **(c)** If the load must rise 0.40 m, how far must the effort move?

Efficiency Problems

24. A pulley system does 450 J of useful work on a load while the worker inputs 600 J. **(a)** What is the efficiency? **(b)** How much energy is wasted to friction?
25. A screw jack has $IMA = 120$ and $efficiency = 25\%$. **(a)** Find the AMA. **(b)** A worker applies 40.0 N to the handle. What force does the jack exert on the car? **(c)** How much energy is dissipated as heat for every 1000 J of work input?
26. A compound machine has three components: a lever (efficiency 95%), a pulley system (efficiency 80%), and an inclined plane (efficiency 60%). Find the total efficiency of the compound machine.
27. A worker uses a ramp (length 5.0 m, height 1.0 m) to roll a 300 kg barrel into a truck. She exerts 800 N along the ramp. **(a)** Calculate W_{in} and W_{out} . **(b)** Find the efficiency. **(c)** Find the IMA and AMA. **(d)** Calculate the friction force on the ramp.

28. A motor with input power 2.0 kW drives a machine that delivers 1.6 kW of useful output power. **(a)** What is the efficiency? **(b)** How much power is dissipated as heat? **(c)** If the motor runs for 30 minutes, how much energy is wasted?

Machine Analysis Exercises

29. Classify each of the following as a lever (and state its class), pulley, inclined plane, wedge, screw, or wheel and axle: **(a)** a zipper, **(b)** a flagpole pulley, **(c)** a bolt tightened with a wrench, **(d)** a seesaw, **(e)** a doorknob, **(f)** a nail being pulled by a hammer head used as a lever, **(g)** a spiral staircase, **(h)** a bicycle rear derailleur, **(i)** a wheelchair ramp, **(j)** a pair of pliers.
30. For a single movable pulley ($\text{IMA} = 2$), draw the rope arrangement. Show the two rope segments supporting the load and the one segment on which effort is applied. Confirm that $\text{IMA} = 2$ by noting the distances: if the load rises 1 m, how far must you pull the rope?
31. A bottle opener is used to pry off a bottle cap. Identify the fulcrum, the effort point, and the load point. What class of lever is this? Estimate the IMA from the approximate geometry of a typical bottle opener.
32. A pair of scissors has two blades, each acting as a first-class lever sharing the pivot (screw in the center). The effort is applied at the handles (12 cm from the pivot) and the cutting occurs 3 cm from the pivot. **(a)** What is the IMA for cutting? **(b)** If you squeeze the handles with 20 N, what force do the blades exert on the material being cut (ideal)? **(c)** Does cutting near the pivot or near the blade tip require less effort? Explain.
33. A bicycle's chainring has 48 teeth (radius 9.5 cm) and the rear sprocket has 16 teeth (radius 3.2 cm). The rear wheel has radius 34 cm. **(a)** Find the gear ratio (chainring/sprocket). **(b)** When the pedals rotate once (moving the chainring once around), how far does the bicycle move? **(c)** If the cyclist exerts 100 N on the pedals at a crank length of 17 cm, what is the driving force at the road surface (ideal)?

Applied Engineering Problems

34. **(Bridge design)** A drawbridge (mass 4000 kg, length 12.0 m) is raised by cables attached 2.0 m from the hinge end. Model the bridge as a second-class lever with the hinge as the fulcrum. **(a)** Identify the load arm and effort arm. **(b)** What cable tension is required to hold the bridge at 30° from horizontal? (Use the component of weight perpendicular to the bridge as the effective load.)
35. **(Car jack)** A scissor jack for a car has a screw with pitch 3.0 mm. The handle is 25.0 cm long. The car exerts a downward force of 8.0 kN on the jack. If the efficiency is 20%, what force must the user apply to the handle? Is this force reasonable for a person to apply?
36. **(Compound pulley)** A theatrical rigging system uses a compound block and tackle with 8 rope segments to fly a 400 kg set piece. **(a)** Find the IMA. **(b)** If efficiency is 72%, find the AMA. **(c)** What force must the stagehand apply to raise the set piece? **(d)** The set piece must rise 6.0 m. How much rope must the stagehand pull?
37. **(Human forearm as a lever)** The bicep muscle attaches to the forearm bone approximately 5.0 cm from the elbow joint (fulcrum). The hand is 35 cm from the elbow. **(a)** What class of lever is the forearm? **(b)** What is the IMA? **(c)** If the forearm holds a 15 N book, what force must the bicep exert? **(d)** What does this tell you about why muscles fatigue quickly under load?
38. **(Wedge analysis)** An axe blade is 15 cm long (from edge to back) and 5 cm wide at the back. **(a)** Find the IMA. **(b)** If struck with 500 N, what splitting force is applied perpendicular to the wood grain (ideal)? **(c)** The efficiency of a steel-through-wood wedge is about 40%. What is the actual splitting force?

Chapter 18 Checkpoint

Before moving to Chapter 19, confirm you can do each of the following without consulting your notes.

I can...	See Section
Explain why machines cannot create energy, only redistribute it	18.3
Calculate $MA = F_{out}/F_{in}$ and interpret the result	18.4
Compute $IMA = d_{in}/d_{out}$ from machine geometry	18.5
Distinguish IMA from AMA and explain why $AMA \leq IMA$	18.5–18.6
Calculate machine efficiency and the energy lost to friction	18.7
Identify the three classes of levers and compute lever IMA	18.8
Count rope segments to find pulley system IMA	18.9
Compute inclined plane $IMA = L/h$	18.10
Compute screw $IMA = 2\pi r/p$	18.11
Compute wheel-and-axle $IMA = R/r$	18.12
Find total MA and efficiency of a compound machine	18.13
Identify and correct the six most common MA mistakes	18.14

Continue to Chapter 19: Torque and Rotational Motion →